

DESIGNS OF THE FIRST-PHASE BEAMLINES FOR SIAM PHOTON SOURCE-II

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Abstract

Siam photon source-II (SPS-II) is a new synchrotron facility that is going to be built in Thailand. There are seven beamlines to be constructed together with the new machine. These consist of one soft X-ray beamline, two X-ray absorption beamlines, three X-ray scattering beamlines and one imaging beamline in the lineups. The designs and the selections of insertion devices, front end and beamline components will be presented together with the optical simulation results and the considerations for thermal load management using the combination of front-end components, filters, white/pink beam slits and mirrors along each beamline. New experimental station equipment and the existing equipment from the current Thai synchrotron facility (Siam Photon Source-I) that will be transferred to SPS-II will also be discussed.

INTRODUCTION

Siam Photon Source-II (SPS-II) is a new fourth-generation synchrotron project of Thailand. The 14-cell double-triple bend achromat storage ring of SPS-II can accommodate 21 insertion device beamlines and 7 bending magnet beamlines [1]. The ring circumference is 327.6 m and there are one standard and one middle straight section of each cell with 3.81 and 1.53 m flange-to-flange physical space available for insertion device installations.

There are seven beamlines planned to be installed in the first phase of SPS-II operation. The lineups include: one soft X-ray beamline – High-Resolution Soft X-ray Spectroscopy (HRSXS), two X-ray absorption beamlines – Tender X-ray Absorption Spectroscopy (TXAS) and Hard X-ray Absorption Spectroscopy (HXAS), one imaging beamline – X-ray Micro-Computed Tomography (XMCT), and three X-ray scattering beamlines – Small and Wide Angle X-ray Scattering (SWAXS), High-Resolution X-ray Diffraction/Micromolecular Crystallography (HRXRD/MX) and Grazing Incidence X-ray Diffraction/Total-Reflection X-ray Fluorescence (GIXRD/TRXRF). The energy ranges and the selected insertion devices are listed in Table 1.

BEAMLINE CONCEPTUAL DESIGNS

The main photon sources selected for the first-phase beamlines are in-vacuum undulators (IVU). The period length of 18 mm has been selected as a standard parameter for four beamlines. This period length allows the higher energy photon to be sufficiently produced from the 2.0 m total magnetic length while keeping the energy spectrum discontinuity between the first and third harmonic as small

as possible under the constraint that the minimum physical gap is 4.0 mm (defined by the vertical beam stay clear of SPS-II). 23 mm period was selected for TXAS for continuous energy between 1-13 keV. An elliptically polarized undulator with the magnetic period of 58 mm is planned for HRSXS for polarization control, while an in-vacuum wiggler (IVW) with the magnetic period of 50 mm is planned for XMCT that needs high photon energy. Out-vacuum multipole wiggler (MPW) with a magnetic period length of 120 mm (9 periods) is also considered as a cost-effective alternative. The radiation spectra from the selected photon sources are provided in Fig. 1.

Table 1: Beamlines and Insertion Devices of SPS-II

Beamline	Energy (keV)	Insertion device
HRSXS	0.9-2.5	EPU 58 1.6 m
TXAS	1-13	IVU 23 0.9 m
HXAS	6-35	IVU 18 2.0 m
XMCT	10-60	IVW 50 2.0 m
SWAXS	8-20	IVU 18 2.0 m
HRXRD/MX	8-25	IVU 18 2.0 m
GIXRD/TRXRF	8-28	IVU 18 2.0 m

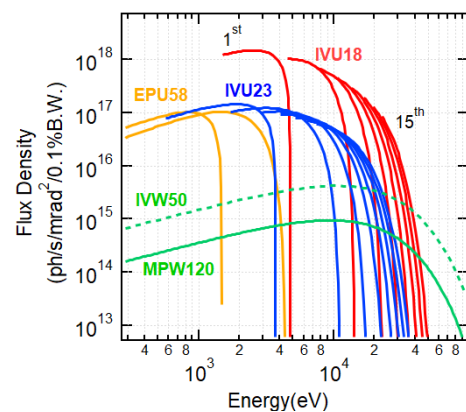


Figure 1: Flux density calculation from the insertion devices of SPS-II.

The conceptual design of the front ends of SPS-II (Fig. 2) is based on the design used at Taiwan Photon Source [2]. The components are standardized for all the seven beamlines. The high heat-load components listed from upstream to downstream are 1) Pre-mask, 2) Fixed Mask, 3) Photon Absorber and 4) X-Y Slits which consist of two motorized L-shape slit blocks. There are two beam positioning monitors (BPMs) placed around 5.5 m away

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from each other. Redundant heavy metal shutters are located at the downstream end next to the ratchet wall. There is a space available for a clearing magnet between the BPMs in case that it needs to be installed in the front end later.

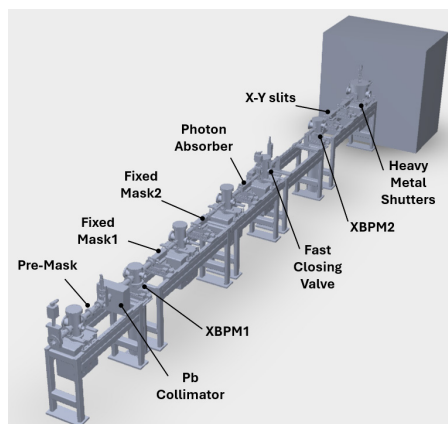


Figure 2: A front end of SPS-II.

The optical design of HRSXS is based on SX-700 plane grating monochromator (250 and 1200 lines/mm). The beamline is split into two branches in time-sharing mode with two toroidal mirrors. The beam focusing is made using a toroidal mirror for the outboard branch, and a K-B mirror system for the inboard branch. From Shadow ray tracing [3], the spot sizes at 1 keV with the exit slit setting of 20 μm are 54 $\mu\text{m} \times 12 \mu\text{m}$ and 35 $\mu\text{m} \times 4 \mu\text{m}$ for the toroidal focusing and K-B focusing, respectively. The flux is in the order of 10^{11} photons/sec for the whole energy range.

TXAS and HXAS share similar focusing concept where the first mirror is used for beam collimating and then after the monochromator, two mirrors are used for independent horizontal and vertical focusing to the sample position. TXAS is a windowless beamline that utilizes a double-crystal monochromator (DCM) with InSb(111) and Si(111) crystal pairs for 1.7-13 keV. The design also includes a second monochromator for lower energy range down to 1 keV. As this may require heat-sensitive crystals such as beryl and KTP, it is important to design a heat load filter in the white beam section of TXAS. A fixed-exit higher harmonic rejection system is planned to be placed between the collimating mirror and the DCMs. The cut-off energies of the low-pass filtering Rh coated mirrors at different angles are shown in Fig. 3. The focused beam size at the sample position at 13 keV is 8 $\mu\text{m} \times 4 \mu\text{m}$ from Shadow. The simulated flux is between 10^{11} - 10^{13} photons/sec over the entire 1-13 keV range. HXAS uses a combination of diamond window, filters and white beam slit to minimize the heat load to the channel-cut monochromator containing Si(111) and Si(311). This type of monochromator is selected to allow for rapid sweeping of photon energy for quick-X-ray absorption fine structure measurements with a goal of 50 Hz repetition rate. The focusing mirrors of HXAS contain multiple metal stripes which can be selected to cut off higher harmonics of the beam after the monochromator without the need for dedicated higher harmonic rejection

systems. The focused beam size at the sample position at 13 keV is 72 $\mu\text{m} \times 11 \mu\text{m}$ from Shadow (0.15 μrad RMS slope error is assumed for all mirrors). The simulated flux is between 10^{11} - 10^{13} photons/sec over the entire 6-35 keV range.

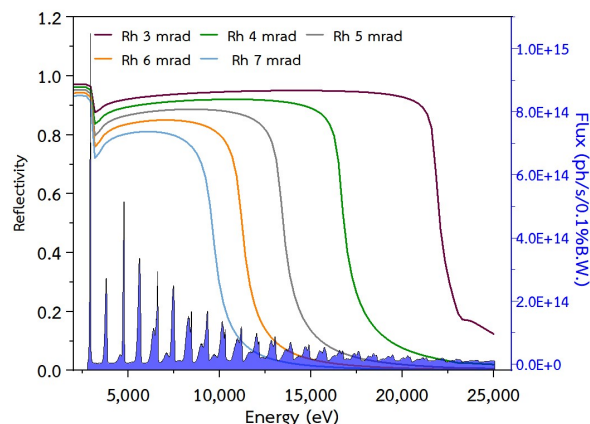


Figure 3: Reflectivity curves for Rh coated on a mirror with grazing angle between 3 and 7 mrad in comparison with the flux coming from IVU23 at 6 mm magnetic gap.

XMCT has no focusing optics and is designed to operate with white-beam and monochromatic beam modes. A water-cooled double multi-layer monochromator (DMM) made with 2 nm-period W/B₄C multilayer was selected for high photon flux output. As the total heat load from IVW50 can be as high as 15 kW, it is important to cut down some power especially at the lower energy by a series of water-cooled filters upstream before DMM. Preliminary simulations showed that CVD diamond filters work well to limit the power and thus the thermal deformation of the first multilayer mirror as shown in Fig. 4. The thermal simulation of 500 μm thick diamond filter is shown in Fig. 5. The design goal of the beamline is to achieve 10^{13} photons/sec monochromatic beam at the sample position under the operation with suitable high heat load filters in the beam path.

All scattering beamlines use identical white beam conditioning systems and DCMs with Si(111) that cover the energy selection between 5-28 keV. HRXRD station require no focusing so it sits right after the DCM. SWAXS and GIXRD/TRXRF share the same optical layout with 2:1 demagnification ratio, while HRXRD/MX needs the demagnification ratio of approximately 26:1 vertically and 172:1 horizontally for MX. These are made using K-B focusing from the primary sources. The focused beam size at 8 keV is 109 $\mu\text{m} \times 62 \mu\text{m}$ and the flux between 4×10^{10} - 3×10^{13} photons/sec over the entire energy range for SWAXS and GIXRD/TRXRF (with 0.20 μrad RMS slope error). MX end station of HRXRD/MX has the spot size of 1.0 $\mu\text{m} \times 0.6 \mu\text{m}$ at the flux of 5×10^{12} photons/sec at 12 keV (with 0.10 μrad RMS slope error and beam defining slits).

Si crystals in all DCMs are cooled by liquid nitrogen supplied by a central storage and distribution lines installed along the ratchet wall. In the beamline design concept, the heat load at cryo-cooled DCMs is limited to around 100 W. This can be made by using water cooled heat load

filters/windows and by limiting the opening angles of the white beam slits to partially cut off low-energy photons emitted from the source at large emission angle (Fig. 6). It is estimated that the consumption of liquid nitrogen will be less than 40 l/hour during the user operation in the first phase.

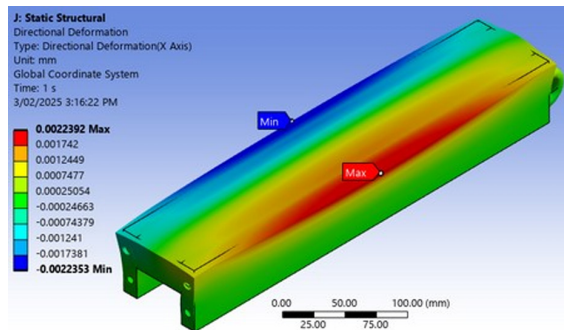


Figure 4: Surface deformation of Si substrate of a multi-layer monochromator.

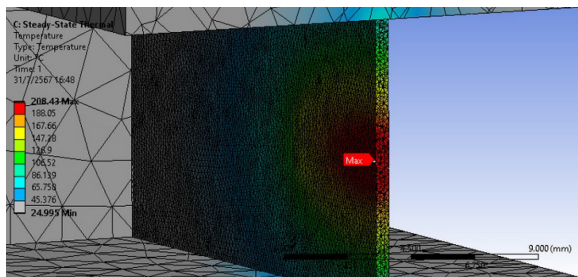


Figure 5: Heat load simulation on a diamond filter.

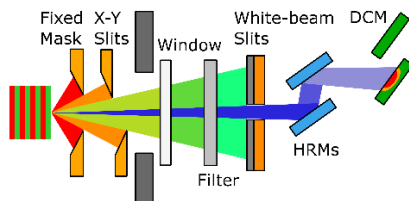


Figure 6: A schematic diagram of heat load sharing and LN2 cooling of the monochromator.

END STATIONS

Most of the end stations that are currently in-operation at SPS-I will be transferred to SPS-II. The list of end station equipment is provided in Table 2. These are detectors that are expected to perform much better with the greatly improved flux from SPS-II.

For scattering and diffraction experimental setups, the technology of area hybrid photon counting detectors are expected to change quickly. Higher photon count rate and faster frame rate are expected to be common features in the near future. The procurement schedule for the detector is set to be near the project completion to ensure that SPS-II will be able to acquire the best cost/performance options for the first service operation time. Planning for an infrastructure capable of handling and transferring large data for SPS-II will also be an important part of the project. The 2D detector for SWAXS will be operated inside a 12 m long

vacuum flight tube. It is expected that the detector will be an in-vacuum type.

Table 2: Equipment to be used at SPS-II Experimental Stations

Equipment	SPS-II end stations
Scienta R4000	HRSXS
Elmitec SPELEEM-III	HRSXS
RaySpec Si drift detectors (4, 7, and 13 element types)	HRSXS, TXAS, HXAS, GIXRD/TRXRF
Dectris Mythen 5k	HRXRD (upgraded to 20 1k modules)
Dectris Pilatus 3 1M	GIXRD/TRXRF

HUTCHES AND UTILITIES

First optical enclosures (FOEs) of all beamlines (Fig. 7) will be made with the shield walls consisting of steel/Pb/steel sheets. It is expected that 40 mm Pb will be used for the end walls and 6 mm Pb for the side walls and roofs. This thickness will required sufficient local shielding by Pb collimators and Bremsstrahlung blocks. The detailed design of the local shielding will be performed with simulation programs such as FLUKA or PHITS. Experimental hutches for all hard X-ray end stations, except for XMCT which uses white beam mode, will be made by sufficiently thick steel sheets without Pb.

The interlock systems for equipment protection (EPS) and personal safety (PSS) will be isolated from each other. It is expected that a reliable ethernet-based field bus such as EtherCAT will be utilized for EPS system to help with cabling installation and maintenance tasks. All sensors of PSS are expected to be hard-wired. They must be redundant and diversified.

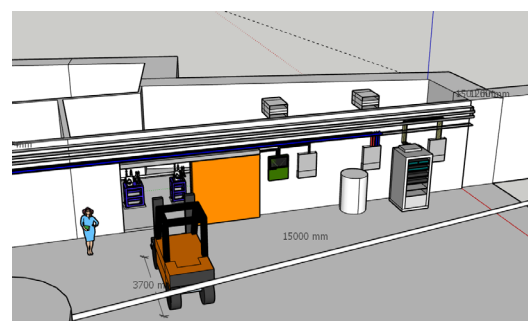


Figure 7: Conceptual design of hutches.

CONCLUSION

The summary of the new insertion device beamlines of SPS-II, a new synchrotron facility in Thailand, are presented. The conceptual design and plans for the first-phase beamlines together with details related to front-end components, optical designs for various beamlines, and plans for infrastructure, including hutches and the transfer of existing equipment from the SPS-I facility.

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